

MAR 580: Advanced Population Modeling

Fall 2025

Production modeling Lab, performance testing

The Pella-Tomlinson production model dynamics can be described by the equation:

$$\ln(B_{y+1}) = \ln(B_y + B_y * (r/(m-1)) * \left(1 - \left(\frac{B_y}{K}\right)^{m-1}\right) - u_y B_y) + \eta_y$$

where B_t is the biomass in year y , r is the intrinsic growth rate, K is the carrying capacity, and $u_y B_y = C_y$ is the predicted catch for year y .

$m > 1$ is the shape parameter that determines the location of B_{MSY} .

We assume that the stock is in an unfished state in the initial (1935) year (i.e. $B_{1935} = K$).

As for our Schaefer model, we assume the survey index observations are proportional to biomass, with the logs of the index being normally distributed with estimated observation error variance. Assume the logs of the catches are also normally distributed with a standard deviation of 0.05. The observation likelihood function is then:

$$L(\theta|D) = \prod_y \frac{1}{\sqrt{2\pi}\sigma} \exp^{-\frac{[\ln I_y - \ln \hat{I}_y]^2}{2\sigma^2}} * \prod_y \frac{1}{0.05\sqrt{2\pi}} \exp^{-\frac{[\ln C_y - \ln(u_y B_y)]^2}{2(0.05)^2}}$$

Tasks:

1. **BONUS** Derive a maximum likelihood estimate for σ , the observation error standard deviation.
2. Fit the Pella-Tomlinson model to the survey and catch data for *Pleuronectes electronica* (include the analytical MLE for q (& σ too if you complete #1) in your model). Evaluate model fit, interpret results, report parameter estimates, and provide recommendations for fishery management based on the results.
3. Compare the results of the P-T model to those of the Schaefer model, determine a best model if there is one, and comment on how multiple models could be considered in management advice.
4. Based on the evaluation of model results in #2, provide rationale for possible changes to the model (additional / reduced complexity), and fit an additional alternative model based on these conclusions. Comment on the evidence for this model over the P-T given the available data.
5. Use the fitted P-T model to generate 500 new data sets based on the best estimates (hint: use the 'SIMULATE' functionality in TMB). Fit both the P-T and Schaefer models to each of these data sets, and summarize the distributions of relative errors for r , K , and the terminal year biomass and exploitation rates. Compute the frequency for which the P-T model is selected by AIC as the 'best' model. Compare the results of the performance testing to the conclusions about the models made in #3.
6. **BONUS BONUS** For the initial P-T model, compute a likelihood profile for the shape parameter (m), and show the associated uncertainty in the estimate for this parameter using an approximate 95% confidence interval.